Dissertation Problem, Purpose, Significance (PPS) Presentation

In partial fulfillment of the degree Doctor of Business Administration
Walsh College

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Dissertation Problem Statement - Official Slide

The general problem is it is not known how reliance on carbon-intensive grid electricity impacts the financial and environmental sustainability of data center operations (Cushman & Wakefield, 2024; MeitY, 2024).

The specific problem is it is not known how wind-solar hybrid energy systems influence the reliability of continuous power delivery for Tier III data centers in Bangalore and Chennai. (Jain & Gupta, 2023; MNRE, 2024).

Approvals:____________________ (DPD) ________________________(CAO)

Dissertation Purpose Statement - Official Slide

The purpose of this non-experimental quantitative correlational study is to examine the relationship between district-specific wind-solar hybrid energy variables (e.g., solar irradiance and wind speed) and the reliability of continuous power supply for Tier III data centers in Bangalore and Chennai.

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Dissertation Significance of the Study - Official Slide

The significance of this study is that it may contribute to business practice by providing data center operators, infrastructure developers, and energy regulators with insights into the relationship between hybrid renewable energy variables and power supply reliability. Additionally, this study may contribute to the existing body of knowledge by addressing the gap in empirical research on the effectiveness of wind-solar hybrid systems in supporting continuous operations in high-density urban data center environments. The findings may also support more sustainable and cost-efficient energy strategies, potentially reducing reliance on carbon-intensive grid electricity and improving long-term operational resilience.

Approvals:____________________ (DPD) ________________________(CAO)

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References

Cushman & Wakefield. (2024). *Data center outlook: India's digital evolution and the energy challenge*. https://www.cushmanwakefield.com/en/india/insights/data-centre-report-2024

Jain, S., & Gupta, R. (2023). Temporal complementarity of wind and solar energy resources in the Indian subcontinent. *Journal of Renewable and Sustainable Energy*, *15*(2), Article 023038. https://doi.org/10.1063/5.0123456

Gartner. (2025). *Sustainable infrastructure: Navigating the energy consumption of AI-driven data centers*. Gartner Research.

Solar Irradiance data sets: https://iced.niti.gov.in/energy/fuel-sources/solar/irradiance

Wind Speed + Direction data sets: https://iced.niti.gov.in/energy/fuel-sources/wind/speed

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